

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

550 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

550 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.99 Cr after input tax credit, generating 8,43,150 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹4.02 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	550 kWp
Year 1 generation	8,43,150 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹60.37 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.09 Cr
GST at 5% — recovered as ITC	₹9.95 L
Net capitalised cost	₹1.99 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹50.10 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,99,04,762	79,61,905	1,19,42,857	20,03,852
2	1,19,42,857	47,77,143	71,65,714	12,02,311
3	71,65,715	28,66,286	42,99,429	7,21,387
4	42,99,428	17,19,771	25,79,657	4,32,832
5	25,79,657	10,31,863	15,47,794	2,59,699
6	15,47,795	6,19,118	9,28,677	1,55,820
7	9,28,677	3,71,471	5,57,206	93,492
8	5,57,206	2,22,882	3,34,324	56,095
9	3,34,323	1,33,729	2,00,594	33,657
10	2,00,594	80,238	1,20,356	20,194
11	1,20,357	48,143	72,214	12,117
12	72,214	28,886	43,328	7,270
Tax shield, years 1-12				49,98,726

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

550 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	8,43,150	60,36,954	—	79,61,905	(4,84,472)	65,21,426	65,21,426
2	8,38,934	62,47,040	3,30,000	47,77,143	2,86,889	56,30,151	1,21,51,577
3	8,34,740	64,64,437	3,30,000	28,66,286	8,22,528	53,11,909	1,74,63,486
4	8,30,566	66,89,399	3,30,000	17,19,771	11,67,702	51,91,698	2,26,55,184
5	8,26,413	69,22,190	3,30,000	10,31,863	13,99,423	51,92,767	2,78,47,951
6	8,22,281	71,63,083	3,30,000	6,19,118	15,63,931	52,69,152	3,31,17,103
7	8,18,170	74,12,358	3,30,000	3,71,471	16,88,996	53,93,362	3,85,10,465
8	8,14,079	76,70,308	3,30,000	2,22,882	17,91,314	55,48,994	4,40,59,459
9	8,10,008	79,37,235	3,30,000	1,33,729	18,80,932	57,26,303	4,97,85,762
10	8,05,958	82,13,451	3,30,000	80,238	19,63,913	59,19,538	5,57,05,300
11	8,01,929	84,99,279	3,30,000	48,143	20,43,928	61,25,351	6,18,30,651
12	7,97,919	87,95,054	3,30,000	28,886	21,23,215	63,41,839	6,81,72,490
13	7,93,929	91,01,121	3,30,000	17,331	22,03,154	65,67,968	7,47,40,458
14	7,89,960	94,17,840	3,30,000	10,399	22,84,611	68,03,230	8,15,43,688
15	7,86,010	97,45,581	3,30,000	6,239	23,68,143	70,47,438	8,85,91,126
16	7,82,080	1,00,84,727	3,30,000	3,744	24,54,128	73,00,600	9,58,91,726
17	7,78,169	1,04,35,676	3,30,000	2,246	25,42,831	75,62,845	10,34,54,571
18	7,74,279	1,07,98,838	3,30,000	1,348	26,34,458	78,34,380	11,12,88,951
19	7,70,407	1,11,74,637	3,30,000	809	27,29,175	81,15,462	11,94,04,413
20	7,66,555	1,15,63,514	3,30,000	485	28,27,129	84,06,386	12,78,10,799
21	7,62,722	1,19,65,925	3,30,000	291	29,28,456	87,07,468	13,65,18,267
22	7,58,909	1,23,82,339	3,30,000	175	30,33,289	90,19,050	14,55,37,317
23	7,55,114	1,28,13,244	3,30,000	105	31,41,757	93,41,488	15,48,78,805
24	7,51,339	1,32,59,145	3,30,000	63	32,53,991	96,75,154	16,45,53,959
25	7,47,582	1,37,20,563	3,30,000	38	33,70,128	1,00,20,436	17,45,74,395
26	7,43,844	1,41,98,039	3,30,000	23	34,90,302	1,03,77,737	18,49,52,132
27	7,40,125	1,46,92,131	3,30,000	14	36,14,658	1,07,47,473	19,56,99,605
28	7,36,424	1,52,03,417	3,30,000	8	37,43,340	1,11,30,077	20,68,29,682
29	7,32,742	1,57,32,496	3,30,000	5	38,76,499	1,15,25,997	21,83,55,679
30	7,29,078	1,62,79,987	3,30,000	3	40,14,292	1,19,35,695	23,02,91,374
Total	2,35,43,415	31,06,20,008	95,70,000	1,99,04,761	7,07,58,640	23,02,91,374	23,02,91,374

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

550 kWp

PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹4.02 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹23.03 Cr
30-year generation	2,35,43,414 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs **₹2.98** across its life against **₹7.65** bought from APSPDCL today — and the grid figure reaches **₹23.86** by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,35,43,414 units, avoiding 19,305.6 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.09 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 550 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.